Year 13 Maths - Projectiles

Similar questions - Time of flight, range and clearing obstacles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A ball is projected from a point O on horizontal ground with initial velocity components 20 $\mathrm{m\,s^{-1}}$ horizontally and 15 $\mathrm{m\,s^{-1}}$ vertically. Model the ball as a particle moving freely under gravity, taking $g = 10$.

(a) Find the time of flight and the horizontal distance from O to the point where the ball lands. **[3]**

(b) A vertical wall stands 40 m from O. Find the height at which the ball passes over the wall. **[3]**

(c) Find the speed of the ball as it passes over the wall. **[3]**

2 A small ball B is projected from a point O on horizontal ground with velocity components U $\mathrm{m\,s^{-1}}$ horizontally and V $\mathrm{m\,s^{-1}}$ vertically. B moves freely under gravity.

(a) Show that the path of B has equation $2U^2y = 2UVx - gx^2$. **[3]**

B just clears a vertical wall of height $\frac{1}{3}a$ m at a horizontal distance a m from O, and lands on the ground at a horizontal distance $2a$ m beyond the wall.

(b) Show that $UV = \frac{3}{2}ga$ and $U^2 = 3ga$. **[4]**

(c) Find the angle of projection, to 3 significant figures. **[2]**

(d) Given that the speed of projection is 30 $\mathrm{m\,s^{-1}}$ and $g = 10$, find a, and hence the greatest height of B above the ground. **[4]**

Question 1

(a) Returns to the ground when $15t - 5t^2 = 0$: $T = 3$ s. Range $= 20 \times 3 = 60$ m

(b) At $x = 40$, $t = 2$: $y = 15 \times 2 - 5 \times 4 = 10$ m

(c) $v_y = 15 - 10 \times 2 = -5$, speed $= \sqrt{20^2 + 5^2} = \sqrt{425} = 20.6\ \text{m s}^{-1}$

Question 2

(a) $x = Ut$ and $y = Vt - \frac{1}{2}gt^2$. Eliminating $t = x/U$: $y = \dfrac{Vx}{U} - \dfrac{gx^2}{2U^2}$, so $2U^2y = 2UVx - gx^2$

(b) Lands at $x = 3a$ when $y = 0$: $2UV \cdot 3a = g(3a)^2$, so $UV = \frac{3}{2}ga$. At $x = a$, $y = \frac{a}{3}$: $\frac{2}{3}U^2a = 2UVa - ga^2 = 3ga^2 - ga^2 = 2ga^2$, so $U^2 = 3ga$

(c) $\tan\theta = \dfrac{V}{U} = \dfrac{UV}{U^2} = \dfrac{3ga/2}{3ga} = \frac{1}{2}$, so $\theta = 26.6^\circ$

(d) $V^2 = \dfrac{(UV)^2}{U^2} = \dfrac{9g^2a^2/4}{3ga} = \frac{3}{4}ga$. So $U^2 + V^2 = \frac{15}{4}ga = 900$, giving $a = \dfrac{900 \times 4}{15 \times 10} = 24$ m. Greatest height $= \dfrac{V^2}{2g} = \dfrac{3a}{8} = 9$ m